

Q1. What is a string in Python?

= A string is a sequence of characters enclosed with single (' ') or double (" ") quotes.

Example - name = "Akash".

Q2. Which symbol is used to create a tuple?

= Parentheses ( ) are used to create a tuple.

Ex = t = (10, 20, 30).

Q3. Which Python data type stores unique value?

= Set stores unique values.

Q4. Write the syntax to create an empty dictionary.

=> d = { }

Q5. Which loop is generally used to iterate over a sequence?

= for loop.

Q6. Which keyword is used to exit a loop immediately?

=> break

Q7. Can a list contain duplicate values?

= Yes.

Q8. Can a tuple be modified after it is created?

Ans = NO

Q9. What is the index of the first element of a python list.

= 0

10) Which function is used to find the number of elements in a string, list, tuple, set, or dictionary?

Ans. len() function.

11) Differentiate between list and Tuple.

= List

Tuple

① Create using []  
 ② list is mutable  
 ③ Can be modified  
 ④ Ex - [1, 2, 3]

Create using ()  
 Tuple is immutable  
 Cannot be modified  
 Ex = (1, 2, 3)

12) Differentiate between Set and Dic

= Set

Store Unique value

Created using {values}

Duplicate value are removed

Ex - {10, 20}

Dictionary

Store key-value pair

Created using {key: value}

Keys must be unique.

Ex = {"name": "Akash"}

Q13. Write a python program to count the number of characters in a string.

```
= text = input("Enter a string:")  
Count = len(text)  
print("Number of characters:", Count)
```

Q14. Write a python program using a for loop to print number from 1 to 10.

```
= for i in range(1, 11):  
    print(i)
```

Q 15. Write a python program using a while loop to print number from 10 to 1.

```
i = 10
while i >= 1:
    print(i)
    i = i - 1
```

Q 16. String - 5 marks.

Consider text = "Python Programming".  
Write python statements to:

- Print the complete string
- Print the first character
- Print the last character
- Print "Python" using Slicing
- Count the number of character in the string.

⇒ a) Print complete string  
print(text)

⑥ Print first character  
print(text[0])

⑦ Print last character  
print(text[-1])

d) Print "Python" using slicing  
 Print (text [0:6])

e. Count number of Character.  
 Print (len (text))

Q 17. Create the following list.  
 Numbers = [10, 20, 30, 40, 50].

Write Python Statements to:

- Add 60 to the list
- Remove 20
- Change 30 to 35
- Print the first three elements.
- Print the length of the list.

= numbers = [10, 20, 30, 40, 50]

- Add 60  
 number.append (60)  $\Rightarrow$  [10, 20, 30, 40, 50, 60]
- Remove 20  
 number.remove (20)  $\Rightarrow$  [10, 30, 40, 50, 60]
- Change 30 to 35  
 number [1] = 35  $\Rightarrow$  [10, 35, 40, 50, 60]
- Print First three elements  $\Rightarrow$  [10, 35, 40]  
 Print (number [:3])

e. Print length.  
Print (len (numbers))

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Q 18. Create the following tuple:

Fruits ("Apple", "Banana", "Mango", "Orange", "Grapes")

Write Python Statement.

- Print the tuple
- Print "Mango"
- Print the first two elements
- Check ~~check~~ whether "Apple" exists in the tuple
- Find the length of the tuple.

= Fruits = ("Apple", "Banana", "Mango", "Orange", "Grapes")

a) Print tuple  
= print (Fruits)

Output.

= ('Apple', 'Banana', 'Mango', 'Orange', 'Grapes')

b) Print Mango  
= print (Fruits [2])

= Mango

c) Print first two elements  
= print (Fruits [:2])

= ('Apple', 'Banana')

d) Check Apple exists.  
= print ("Apple" in fruits)

= True

e) Find length  
= print (len (fruits))

= 5

19. Consider :

numbers = {10, 20, 30, 20, 40, 10, 50}

- What happen its duplicate values in the set.
- Add 60 to the set.
- Remove 30 from the set.
- Check whether 40 exists in the set.
- Find the number of elements in the set.

Ans numbers = {10, 20, 30, 20, 40, 10, 50}

= Duplicate values are automatically removed because a set only stores unique values.

a). Duplicate are automatically removed.

= print (numbers)

b). Add 60

= number.add(60)

c). Remove 30

= number.remove(30)

d). Check 40 exists

Print (40 in numbers)

e. Number of elements  
= `Print (len(numbers))`

Q. 20 Dictionary -

Create the following dictionary :

```
Student = { "name": "Rahul",  
            "age": 22,  
            "Course": "Python",  
            "City": "Delhi" }
```

(a) Print the student's name .

⇒ `Print (Student ["name"])`

(b) Print the student's age .

⇒ `Print (Student ["age"])`

(c) Add "marks" : 85 to the dictionary -

= `Student ["marks"] = 85`

(d) Change the city to "Mumbai" :

⇒ `Student ["City"] = "Mumbai"`

(e) Display all keys of the dictionary

= `Print (Student.keys())`

Q21. Loops :

A. For loop

a. ~~Write~~ Write a python using a for loop to print numbers from 1 to 10.

=> for i in range(1, 11):  
    print(i)

b) Write a python program using for loop to print all even number from 1 to 20.

= For i in range(2, 21, 2):  
    print(i)

Output:- 2, 4, 6, 8, 10, 12, 14, 16, 18, 20.

B While loop.

a) Write a python program a while loop to print number from 1 to 10

= i = 1  
while i <= 10  
    print(i)  
    i = i + 1

d. Write a Python program using a while loop to print number 10 to 1

```
=  
(  
    i = 10  
    while i >= 1:  
        print(i)  
        i = i - 1
```

e. Write a python program using a while loop to calculate sum of number from 1 to 10.

```
=  
    i = 1  
    total = 0  
    while i <= 10:  
        total = total + i  
        i = i + 1
```

```
    print("Sum =", total)
```

Output :

Sum = 55